

SentinelAI: Next-Gen Mission Readiness

The Copilot for Defense Assets

AI-powered predictive maintenance for military fleets.

From Raw Data to Tactical Intelligence

Ingesting multi-sensor telemetry to predict component failure before it impacts the mission.



Repository:
github.com/Aayushh910/910/bob-ai-hackathon-NexGen

Developed by: Team NexGen

The Readiness Gap: Reactive vs. Predictive

Reactive Approach



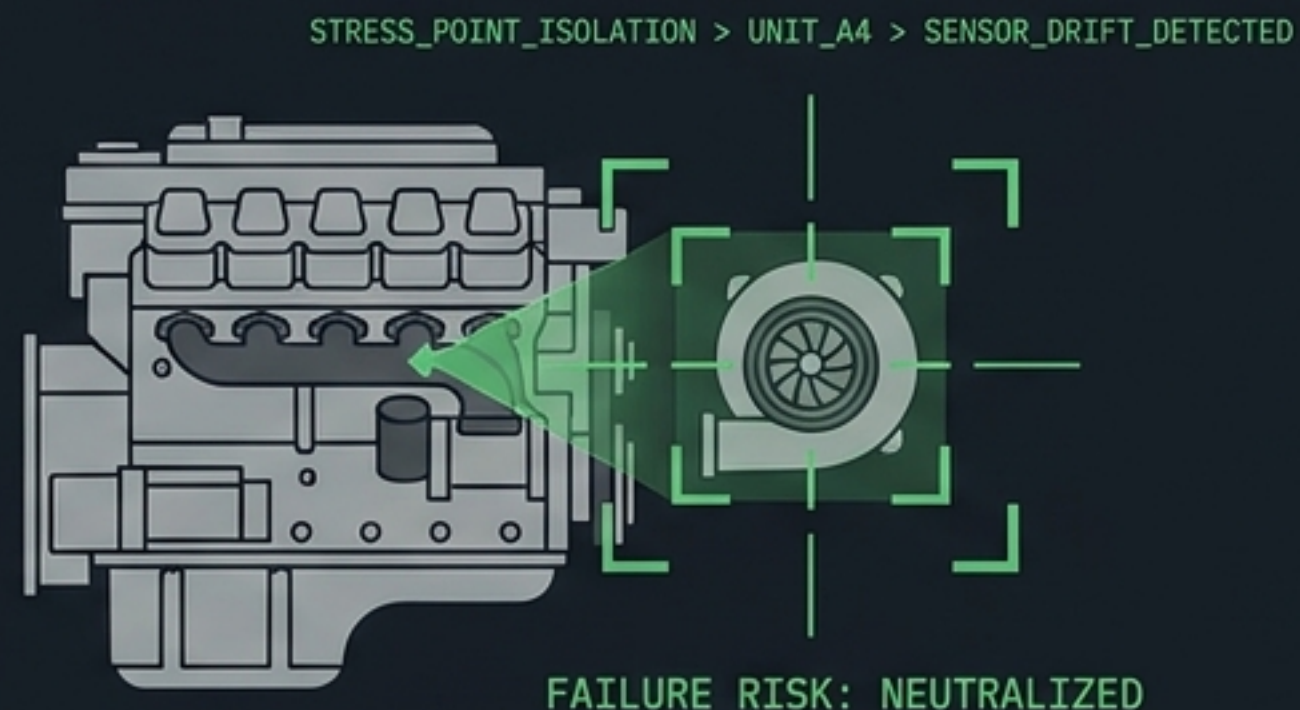
The Current Bottleneck

Maintenance relies on **rigid, fixed schedules** and manual sensor analysis.

The Cost

Unexpected **degradation**, false readiness reporting, and **reduced fleet availability**. 

Predictive Strategy



The SentinelAI Solution

Continuous ingestion of Health and Usage Monitoring System (HUMS) **telemetry**.

The Outcome

Detects **early signs of wear**, translates sensor drift into plain-language commander **commander intelligence**, and **flags risks** before critical failure. 

Tactical Capabilities at a Glance



Component-Specific Predictions

Evaluates exact 50-hour breakdown probabilities for critical subsystems (Engine, Battery, Fuel Pump, Hydraulics).

Deterministic 4-Tier Engine

READY

CAUTION

DEGRADED

NOT_READY

Automatically categorizes fleet assets into actionable states: READY, CAUTION, DEGRADED, or NOT_READY.



Domain-Guarded AI Copilot

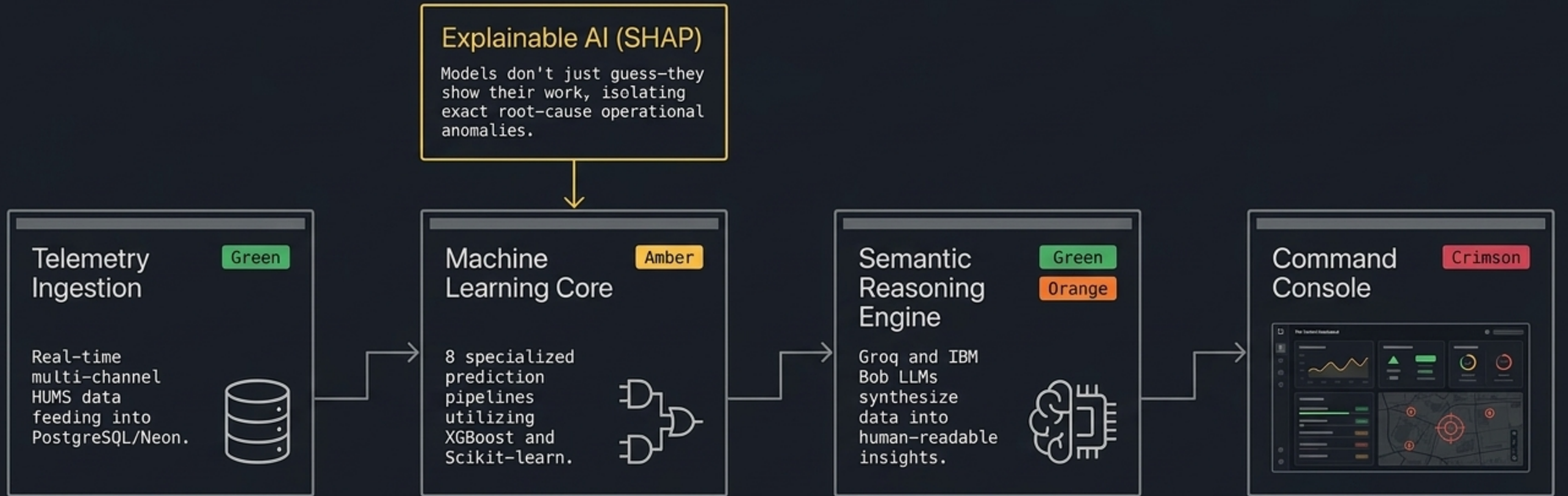
A hybrid conversational AI that understands defense terminology (FMC, NMC, RUL) but strictly blocks out-of-scope queries.



Closed-Loop Work Orders

Automatically ranks maintenance urgency (Critical to Low) and validates recovery once maintenance is completed.

The Data-to-Decision Pipeline



Rapid Deployment Setup

Zero-Friction Launch: Architected for seamless cloud or local deployment.



[STATUS: ONLINE]

```
git clone https://github.com/.../bob-ai-hackathon-NexGen.git
cd src/backend && pip install -r requirements.txt
uvicorn app.main:app --reload
npm run dev
```

1. Clone Source

Pull repository locally.



2. Dependencies

FastAPI backend & React frontend setup.



3. Configure .env

Secure database & API key provisioning.



4. Init Engine

Launch live dashboard.



In Action: The Commander's View

Scenario

Commander queries the AI copilot for direct asset intelligence.

Tactical Chat

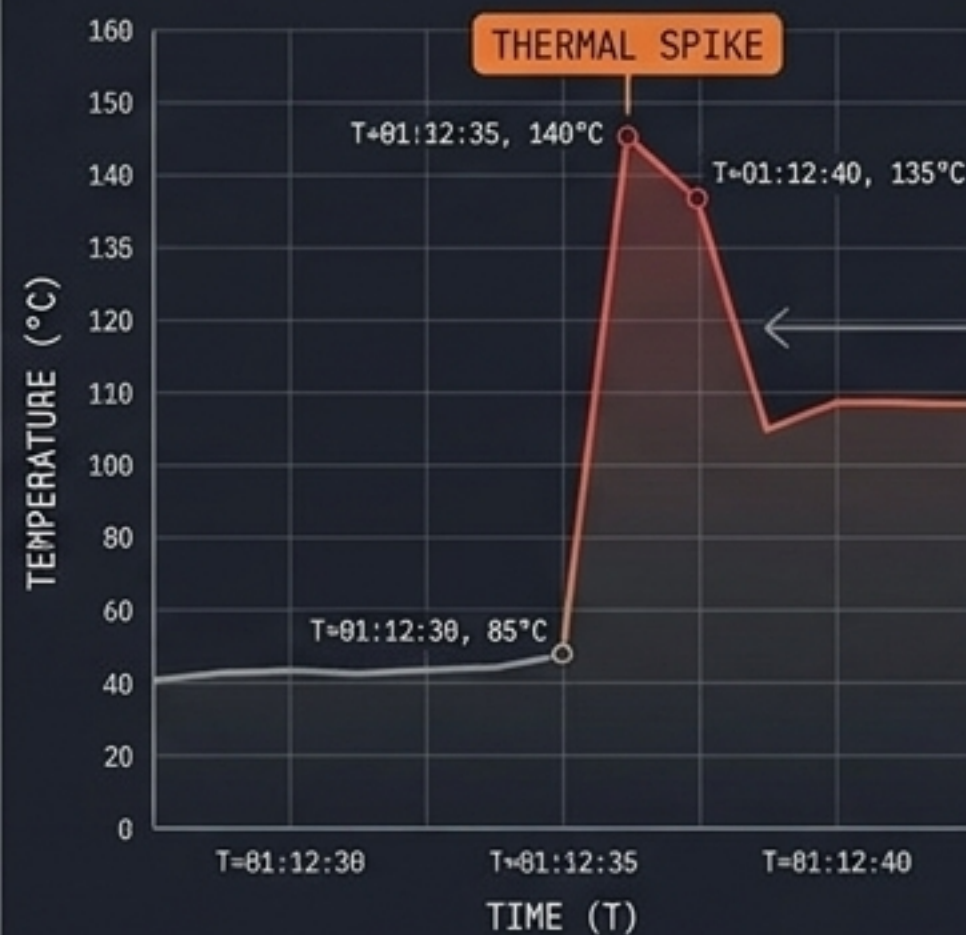
COMMANDER

What is the status of asset A021?

AI COPILOT

DEGRADED

Fuel pump anomaly detected.
Recommend maintenance.
[STATUS: DEGRADED]



Instant Telemetry Retrieval

System pulls live multi-sensor data, identifying a hidden thermal spike.

Root-Cause Identification & Action

Flags asset as DEGRADED via fuel pump anomaly. Automatically generates HIGH priority work order.

Proven Reliability & Scale

20,000+

Data Validation

Telemetry records processed for pinpoint accuracy.

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End-to-End ML Pipelines

Fully serialized, production-ready machine learning pipelines with zero mock data.



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Bulletproof Engineering

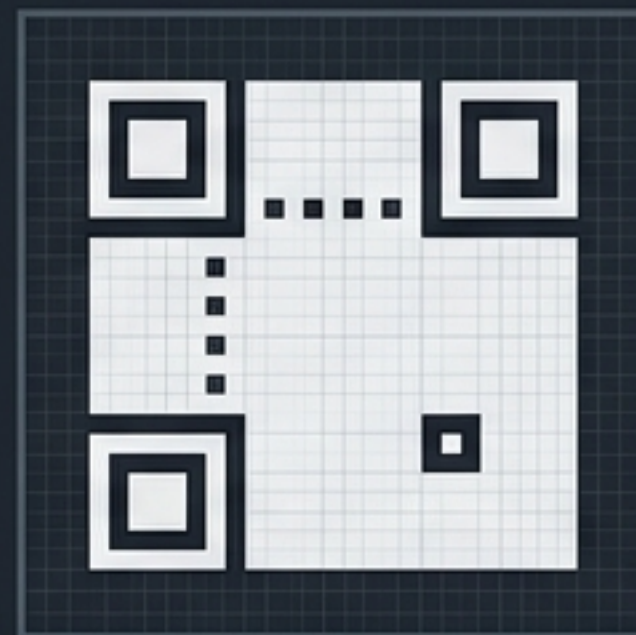
Passing automated tests ensuring mission-critical stability.



Cloud-Native & Secure

Deployed seamlessly on Vercel and Render with secure SMTP/Brevo defense notification engines.

Expanding the Sentinel Ecosystem



Join the Mission

Scan to view the code, read the docs, and **contribute** to the future of defense AI.